# AQA 3.1.12 Acids and bases



Answer in AQA exam style: use the marks as a guide to how much to write and show all working in calculations. Mark your work with the mark scheme at the back. Questions marked as practice questions in the mark scheme were written in AQA style; all others are from AQA past papers.

## Sheet 4 of 8

*20 marks · about 25 minutes*

**Key information**

- Buffer: $[H^+] = K_a \times [HA] \div [A^-]$ — use moles, the volume cancels
- Weak acid + NaOH: **$K_a$ = $[H^+]$ × salt ÷ acid**, where salt = mol NaOH added and acid = mol HA left
- Acid added to a buffer: $A^- + H^+ \rightarrow HA$, so mol HA goes up and mol $A^-$ goes down by the same amount
- $pH = -\log_{10}[H^+]$ and $[H^+] = 10^{-pH}$
- Weak acid on its own: $[H^+] = \sqrt{(K_a \times [HA])}$; $K_a = 10^{-pKa}$

### Section A: Short-answer questions

**01** What is the pH of 0.015 mol $dm^{-3}$ sulfuric acid?

**[1 mark]**

**A** −1.82
**B** −1.52
**C** 1.52
**D** 1.82

**02** State **two** assumptions made when calculating the pH of a weak acid using $[H^+] = \sqrt{(K_a \times [HA])}$.

**[2 marks]**

**03** Which species can behave as a Brønsted–Lowry acid in aqueous solution?

**[1 mark]**

**A** $SO_4^{2-}$
**B** $HCO_3^-$
**C** $BF_3$
**D** $NH_3$

**04** Which is the conjugate base of the Brønsted–Lowry acid $HSO_4^-$?

**[1 mark]**

**A** $H_2SO_4$
**B** $SO_4^{2-}$
**C** $H_3O^+$
**D** $OH^-$

**05** Sodium hydroxide solution is added gradually from a burette to 25 $cm^3$ of 0.080 mol $dm^{-3}$ propanoic acid at 25 °C.

Which indicator is the most appropriate for this titration?

**[1 mark]**

**A** methyl orange (pH range 3.1 – 4.4)
**B** bromothymol blue (pH range 6.0 – 7.6)
**C** cresolphthalein (pH range 8.2 – 9.8)
**D** indigo carmine (pH range 11.6 – 13.0)

## Section B: Longer questions

**06** Propanoic acid ($C_2H_5COOH$) is a weak acid. The acid dissociation constant ($K_a$) for propanoic acid is $1.35 \times 10^{-5}$ mol $dm^{-3}$ at 25 °C

**06.1** Give an expression for the acid dissociation constant for propanoic acid.

**[1 mark]**

**06.2** A student dilutes 25.0 $cm^3$ of 0.500 mol $dm^{-3}$ propanoic acid by adding water until the total volume is 100.0 $cm^3$. Calculate the pH of this diluted solution of propanoic acid. Give your answer to 2 decimal places.

**[4 marks]**

**07** Buffer solution B is made by dissolving 0.656 g of sodium ethanoate ($M_r$ = 82.0) in 50 $cm^3$ of 0.120 mol $dm^{-3}$ ethanoic acid. This buffer solution has pH = 4.87

**07.1** Explain, qualitatively, how this buffer solution resists pH change when a small amount of sodium hydroxide solution is added.

**[2 marks]**

**07.2** Calculate the pH change that occurs when 1.50 $cm^3$ of 0.200 mol $dm^{-3}$ HCl(aq) are added to buffer solution B. Give your answer to 2 decimal places. The acid dissociation constant ($K_a$) for ethanoic acid at 25 °C is $1.78 \times 10^{-5}$ mol $dm^{-3}$

**[7 marks]**

# Mark Scheme

*M1, M2… are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.*

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| Question | Marking guidance | Comments | Mark |
|---|---|---|---|
| 01 | **C** | *Source: 7405/3 Nov 2020 Q34* | 1 |
| 02 | $[H^+] = [A^-]$ (dissociation of water is negligible)<br>[HA] at equilibrium = initial [HA] (dissociation of the acid is negligible) | *Source: Practice question (written for this set)* | 2 |
| 03 | **B** | *Source: 7405/3 Jun 2023 Q16* | 1 |
| 04 | **B** | *Source: Practice question (written for this set)* | 1 |
| 05 | **C** | *Source: 7405/1 Jun 2019 Q09.3*<br>Original question is tick-one-box | 1 |
| 06.1 | $K_a = [H^+][C_2H_5COO^-] \div [C_2H_5COOH]$ | *Source: 7405/1 Nov 2020 Q04.2–04.3*<br>Do not allow ( ) | 1 |
| 06.2 | M1 $[C_2H_5COOH] = 0.125$ mol dm$^{-3}$<br>M2 $[H^+] = \sqrt{(K_a \times [C_2H_5COOH])} = \sqrt{(1.35 \times 10^{-5} \times 0.125)}$<br>M3 $[H^+] = 1.30 \times 10^{-3}$ mol dm$^{-3}$<br>M4 pH = 2.89 | If 0.0125 used lose M1, allow M2, M3 and M4 = 3.39<br>Answer to 2 dp | 4 |
| 07.1 | M1 $OH^-$ ions react with $H^+$<br>M2 the HA ⇌ $H^+ + A^-$ equilibrium shifts right to maintain $[H^+]$<br>**OR** M1 HA + $OH^- \rightarrow H_2O + A^-$; M2 ratio $[CH_3COOH] \div [CH_3COO^-]$ remains roughly constant | *Source: 7405/1 Jun 2025 Q08.5–08.6*<br>M1 allow $OH^-$ ions react with $CH_3COOH$ / HA | 2 |
| 07.2 | M1 n(HA) at start = 0.120 × 50 ÷ 1000 = 0.00600<br>M2 n($A^-$) at start = 0.656 ÷ 82.0 = 0.00800<br>M3 n($H^+$) added = 0.200 × 1.50 ÷ 1000 = 0.000300<br>M4 n(HA) after = 0.00630 **and** n($A^-$) after = 0.00770<br>M5 $[H^+] = 1.78 \times 10^{-5} \times 0.00630 \div 0.00770 = 1.46 \times 10^{-5}$<br>M6 pH = 4.84<br>M7 change = 4.87 – 4.84 = 0.03 | M4 = M1 + M3 and M2 – M3<br>M5 dependent on an attempt at M4<br>M7 answer to 2 dp | 7 |